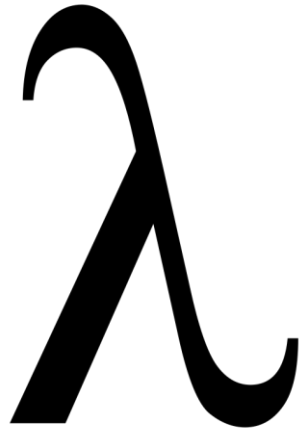




ML Lab 2

Programmazione Funzionale

2025/2026

Università di Trento

Chiara Di Francescomarino



Exercise L2.1

- Write a function `cube` that computes the cube of a real number
- For instance
 - `cube (2.9) = 24.389`



Exercise L2.2

- Write a function `min3` that computes the smallest component of a tuple of type `int*int*int`
- For instance
 - `min3 (2,3,4) = 2`
 - `min3 (3,2,4) = 2`



Exercise L2.3

- Write a function `third` that computes the third element of a list (it doesn't have to work properly on shorter lists).

- For instance,

```
third [2,3,4] = 4
```

```
third [2,3,4,5] = 4
```



Exercise L2.4

- Write a function `reverse` that reverses a tuple of length 3
- For instance,

`reverse (1,2,3) = (3,2,1)`

`reverse (1.0,2,"a") = ("a",2,1.0)`



Exercise L2.5

- Write a function `thirdchar` that finds the string containing the third character of a string (it doesn't have to work properly on shorter strings)
- For instance
`thirdchar "abcd" = "c"`
- You can use support functions



Exercise L2.6

- Write a function `cycle` that cycles a list once, i.e., convert $[a_1, \dots, a_n]$ to $[a_2, \dots, a_n, a_1]$. It doesn't have to work on the empty list.
- For instance
 - `cycle [1,2,3,4] = [2,3,4,1]`
 - `cycle [1] = [1]`



Exercise L2.7

- Write a function `min_max_pair` that, given 3 integers (a tuple), produce a pair consisting of the smallest and the largest value among the 3 integers
- For instance
 - `min_max_pair (1,2,3) = (1,3)`
 - `min_max_pair(3,4,2) = (2,4)`
- You can use support functions



Exercise L2.8

- Write a function `sort` that, given three integers (a tuple), produce a list of them in sorted order
- For instance
 - `sort (1,2,3) = [1,2,3]`
 - `sort (1,3,2) = [1,2,3]`
- You can use support functions



Exercise L2.9

- Write a function `rnd` that rounds a real number to the nearest decimal (10^{th}) digit
- For instance
 - `rnd(2.56) = 2.6`
 - `rnd (5.678) = 5.7`
 - `rnd (3.3) = 3.3`
 - `rnd (4.128) = 4.1`



Exercise L2.10

- Write a function `rem` that, given a list, removes the second element. It doesn't need to work on lists shorter than 2.
- For instance
 - `rem [1,2,3,4] = [1,3,4]`
 - `rem [1,2] = [1]`